Álgebra

1. Leyes de los exponentes

$$a^m a^n = a^{m+n}, \qquad (ab)^m = a^m b^m, \qquad (a^m)^n = a^{mn}, \qquad a^{m/n} = \sqrt[n]{a^m}$$

Si $a \neq 0$,

$$\frac{a^m}{a^n} = a^{m-n}, \qquad a^0 = 1, \qquad a^{-m} = \frac{1}{a^m}.$$

2. Cero La división por cero no está definida.

$$\text{Si } a \neq 0: \quad \frac{0}{a} = 0, \quad a^0 = 1, \quad 0^a = 0$$

$$\text{Para cualquier número } a: \quad a \cdot 0 = 0 \cdot a = 0$$

3. Fracciones

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}, \qquad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \qquad \frac{a/b}{c/d} = \frac{a}{b} \cdot \frac{d}{c}, \qquad \frac{-a}{b} = -\frac{a}{b} = \frac{a}{-b}.$$

4. Teorema del binomio Para cualquier entero positivo n,

$$(a+b)^n = a^n + na^{n-1}b + \frac{n(n-1)}{1 \cdot 2}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3}a^{n-3}b^3 + \cdots + nab^{n-1} + b^n.$$

5. Diferencia de potencias enteras iguales, $n > 1$

$$a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \cdots + ab^{n-2} + b^{n-1})$$

Por ejemplo,

$$a^2 - b^2 = (a-b)(a+b), \quad \text{Diferencia de cuadrados}$$
$$a^3 - b^3 = (a-b)(a^2+ab+b^2), \quad \text{Diferencia de cubos}$$
$$a^4 - b^4 = (a-b)(a^3+a^2b+ab^2+b^3). \quad \text{Diferencia de términos a la 4.}$$

6. Completar el cuadrado Si $a \neq 0$,

$$\left(\frac{\frac{b}{a}}{2}\right) = \frac{b}{2a}$$

$$\left(\frac{b}{2a}\right)^2 = \frac{b^2}{2^2 \cdot a^2} = \frac{b^2}{4a^2}$$

$$\begin{aligned} ax^2+bx+c &= a\left(x^2 + \frac{b}{a}x\right) + c \\ &= a\left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2}\right) + c \\ &= a\left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}\right) + a\left(-\frac{b^2}{4a^2}\right) + c \\ &= a\underbrace{\left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}\right)}_{\text{Esto es } \left(x+\frac{b}{2a}\right)^2.} + \underbrace{c - \frac{b^2}{4a}}_{\text{Llamemos } C \text{ a esta parte.}} \\ &= a\left(x+\frac{b}{2a}\right)^2 + C \\ &= au^2 + C \qquad (u = x + (b/2a)) \end{aligned}$$

7. La fórmula cuadrática Si $a \neq 0$,

$$ax^2 + bx + c = 0$$
$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}.$$

i) $b^2 - 4ac > 0$; $x_1 \neq x_2 \in \mathbb{R}$

ii) $b^2 - 4ac = 0$; $x_1 = x_2 \in \mathbb{R}$

iii) $b^2 - 4ac < 0$; $x_1, x_2 \in \mathbb{C}$

$\alpha \pm j\beta$

$j = \sqrt{-1}$

$x \cdot x = x^2$; $x^2 \cdot x = x^3$

; $x^2 \cdot x^2 = x^4$

$(x-1)(x+1)$ Diferencia de cuadrados

$x \cdot x + x - x - 1 = x^2 - 1$

$(3x)^2 = 3^2 x^2 = 9x^2$; $\left(\frac{1}{2} \cdot x\right)^3 = \left(\frac{1}{2}\right)^3 (x)^3 = \frac{1}{8} x^3$

$\left(\frac{1}{2} \cdot x^2\right)^3 = \left(\frac{1}{2}\right)^3 \cdot (x^2)^3 = \frac{1}{8} \cdot x^{2 \cdot 3} = \frac{1}{8} x^6$

$a^{m/n} = \sqrt[n]{a^m}$

$\sqrt{x} = x^{1/2}$; $x^{1/3} = \sqrt[3]{x}$

$\sqrt[2]{x} = x^{1/2}$; $x^{2/3} = \sqrt[3]{x^2}$

$x^{7/9} = \sqrt[9]{x^7}$

$\frac{1}{x^{1/2}} \cdot \frac{x^{1/2}}{x^{1/2}} = \frac{(1)\, x^{1/2}}{x^{1/2 + 1/2}} = \frac{x^{1/2}}{x} = \frac{\sqrt{x}}{x}$

Racionalizando.

$\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{(1)(2)^{1/2}}{(2)^{1/2}(2)^{1/2}} = \frac{(2)^{1/2}}{2^{1/2+1/2}} = \frac{\sqrt{2}}{2}$

$\frac{1}{x} = x^{-1}$; $\frac{1}{x^2} = x^{-2}$; $\frac{1}{x^3} = x^{-3}$

$\frac{1}{x^{-3}} = x^3$

$$\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd},$$

$$\frac{1}{2}+\frac{2}{3}=\frac{(1)(3)+(2)(2)}{(2)(3)}$$

$$=\frac{3+4}{6}=\frac{7}{6}$$

$$\frac{1}{2}-\frac{2}{3}=\frac{(1)(3)-(2)(2)}{(2)(3)}$$

$$=\frac{3-4}{6}=-\frac{1}{6}$$

$$-\frac{5}{2}+\frac{7}{8}=\frac{(-5)(8)+(7)(2)}{(2)(8)}$$

$$=\frac{-40+14}{16}=\frac{-26}{16}$$

$$=-\frac{13}{8}$$

Extremos y medios

$$\frac{a/b}{c/d}=\frac{a}{b}\cdot\frac{d}{c},$$

$$\frac{\frac{a}{b}}{\frac{c}{d}}=\frac{ad}{bc}$$

Ejemplo: Determinar la gráfica de la función y su respectivo dominio.

$$f(x)=\sqrt{-x^2+5x-6}$$

$$y=\sqrt{-x^2+5x-6}$$

$$y^2=-x^2+5x-6$$

$$y^2+x^2-5x=-6$$

completar cuadrados

$$y^2+(x-5/2)^2-\frac{25}{4}=-6$$

$$y^2+(x-5/2)^2=-6+\frac{25}{4}$$

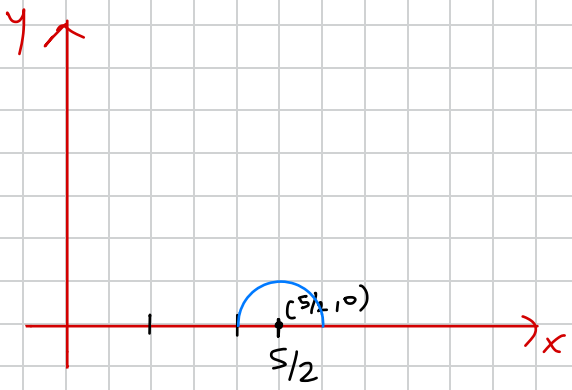
$$y^2+(x-5/2)^2=\frac{-24+25}{4}$$

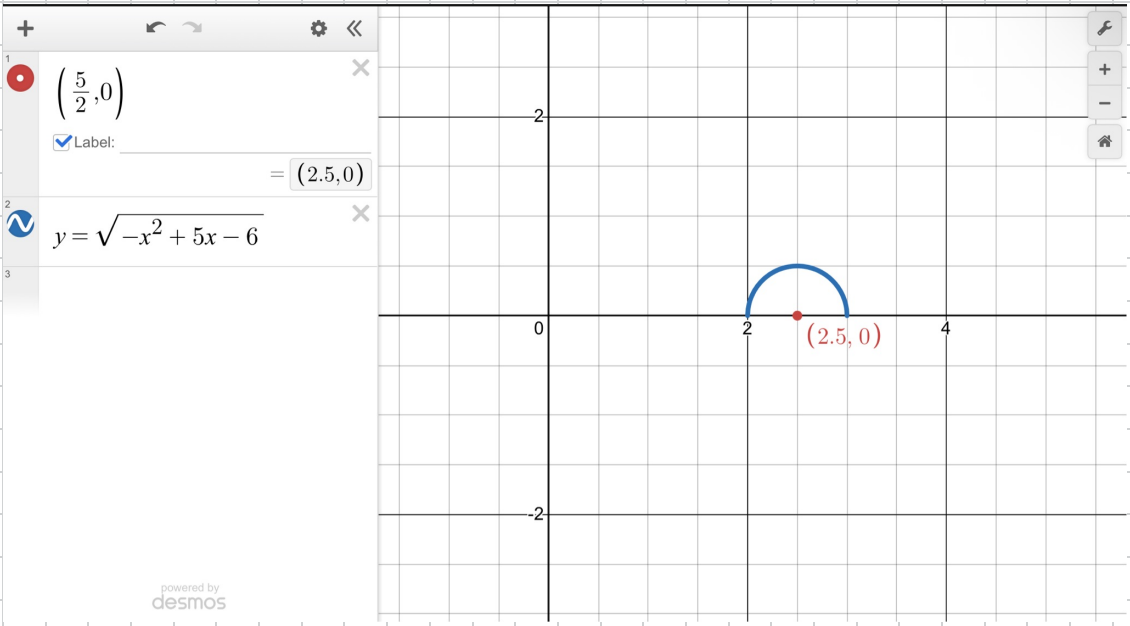
$$y^2+(x-5/2)^2=\frac{1}{4}$$

Circunferencia

centro $(5/2, 0)$

radio $=\frac{1}{2}$





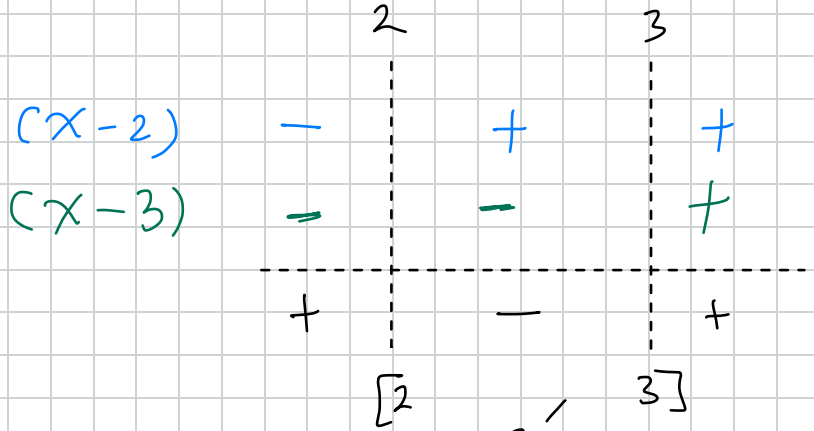
Dominio f: $\{x \mid -x^2+5x-6 \geq 0\}$

$$-x^2+5x-6 \geq 0 \quad (-1)$$

$$x^2-5x+6 \leq 0$$

$$(x-3)(x-2) \leq 0$$

Análisis de Signos.



Dominio f: $\{x \mid x \in [2,3]\} = x \in [2,3]$

$= 2 \leq x \leq 3$